圆锥曲线综合习题册

Conic Sections Comprehensive Exercise Book

一、选择题（Multiple Choice Questions）

1. 圆心是$(2,-3)$，半径是 4 的圆的方程是 (　).The equation of the circle with center $(2,-3)$ and radius 4 is (　).

A. $(x-2)^2+(y+3)^2=4$

B. $\frac{x^2}{16}+\frac{y^2}{9}=1$

C. $(x-2)^2+(y+3)^2=16$

D. $(x+2)^2+(y-3)^2=16$

2. 圆的一般方程是$x^2+y^2-6x+4y-12=0$，它的圆心是(　).The general equation of the circle is $x^2+y^2-6x+4y-12=0$, its center is (　).

A. $(3,-2)$

B. $(-3,2)$

C. $(2,-3)$

D. $(-2,3)$

3. 下列是椭圆方程的是(　).Which of the following is the equation of an ellipse? (　).

A. $\frac{x^2}{5}+\frac{y^2}{10}=1$

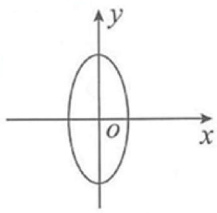
B. $\frac{x^2}{25}-\frac{y^2}{16}=1$

C. $x^2+y^2=16$

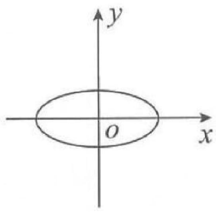
D. $y^2=-20x$

4. 椭圆$\frac{x^2}{100}+\frac{y^2}{36}=1$的焦点坐标是 (　).The focal coordinates of the ellipse $\frac{x^2}{100}+\frac{y^2}{36}=1$ are (　).A. $(\pm 8,0)$　B. $(0,\pm 8)$C. $(\pm 14,0)$　D. $(0,\pm 14)$

5. 椭圆$\frac{x^2}{36}+\frac{y^2}{169}=1$的图像是 (　).The graph of the ellipse $\frac{x^2}{36}+\frac{y^2}{169}=1$ is (　).



A.



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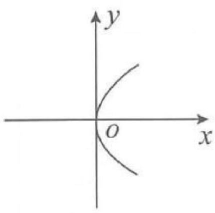

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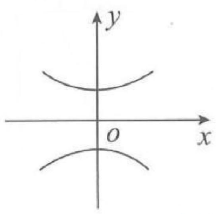


D.

6. 下列是焦点在 y 轴的双曲线的方程是(　).Which of the following is the equation of a hyperbola with foci on the y-axis? (　).

A. $\frac{y^2}{36}-\frac{x^2}{16}=1$

B. $\frac{y^2}{36}+\frac{x^2}{16}=1$

C. $\frac{x^2}{36}-\frac{y^2}{16}=1$

D. $\frac{x^2}{36}+\frac{y^2}{16}=1$

7. 下列是双曲线方程的是 (　).Which of the following is the equation of a hyperbola? (　).

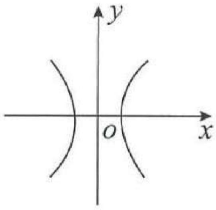
A. $x^2+y^2=25$

B. $x=3y^2$

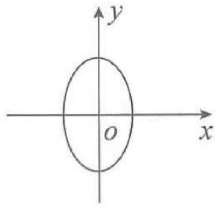
C. $\frac{x^2}{16}+\frac{y^2}{9}=1$

D. $\frac{x^2}{16}-\frac{y^2}{9}=1$

8. $\frac{y^2}{49}-\frac{x^2}{144}=1$的图像是(　).The graph of $\frac{y^2}{49}-\frac{x^2}{144}=1$ is (　).



A.



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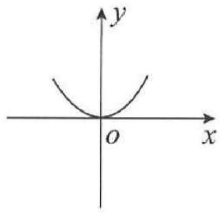

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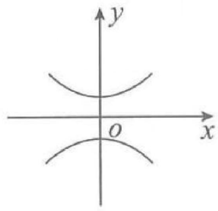


D.

9. 顶点在原点,焦点在 x 轴上,并且经过点$P(3,6)$的抛物线方程是 (　).The equation of the parabola with vertex at the origin, foci on the x-axis, and passing through point $P(3,6)$ is (　).

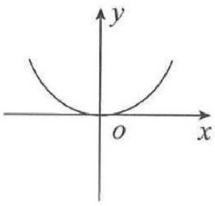
A. $y^2=12x$

B. $y^2=6x$

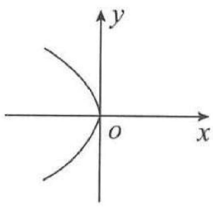
C. $x^2=3y$

D. $x^2=\frac{1}{3}y$

10. $x^2=-8y$的图像是(　).The graph of $x^2=-8y$ is (　).



A.



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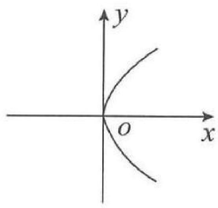

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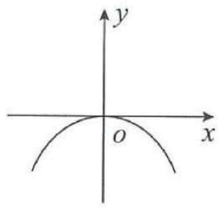
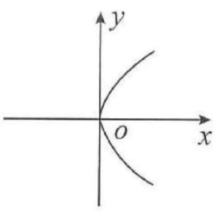
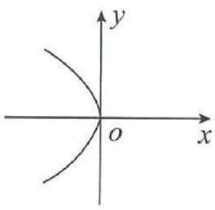


D.

11. $y^2=-20x$的图像是(　).The graph of $y^2=-20x$ is (　).



A.



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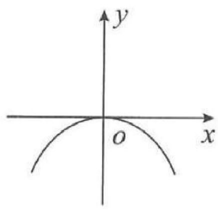

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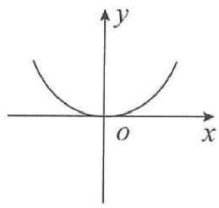

D.

12. 抛物线$y^2=6x$的开口方向是 (　).The opening direction of the parabola $y^2=6x$ is (　).

A. 向上 (Upward)

B. 向下 (Downward)

C. 向左 (To the left)

D. 向右 (To the right)

13. 抛物线$x^2=12y$的开口方向是 (　).The opening direction of the parabola $x^2=12y$ is (　).

A. 向上 (Upward)
B. 向下 (Downward)
C. 向左 (To the left)
D. 向右 (To the right)

14. 焦点在 x 轴上的椭圆方程是 ( ).The equation of the ellipse with foci on the x-axis is ( ).
A. $\frac{x^2}{16}-\frac{y^2}{9}=1$
B. $\frac{x^2}{16}+\frac{y^2}{9}=1$
C. $\frac{x^2}{9}+\frac{y^2}{16}=1$
D. $\frac{y^2}{16}-\frac{x^2}{9}=1$

15. 焦点在 y 轴上的椭圆方程是 ( ).The equation of the ellipse with foci on the y-axis is ( ).
A. $\frac{x^2}{9}+\frac{y^2}{36}=1$
B. $\frac{x^2}{9}-\frac{y^2}{36}=1$
C. $\frac{x^2}{36}+\frac{y^2}{16}=1$
D. $\frac{y^2}{36}-\frac{x^2}{9}=1$

16. 焦点在 x 轴上的双曲线方程是 ( ).The equation of the hyperbola with foci on the x-axis is ( ).
A. $\frac{x^2}{9}+\frac{y^2}{36}=1$
B. $\frac{x^2}{9}-\frac{y^2}{36}=1$
C. $\frac{x^2}{36}+\frac{y^2}{16}=1$
D. $\frac{y^2}{9}-\frac{x^2}{36}=1$

17. 焦点在 y 轴上的双曲线方程是 ( ).The equation of the hyperbola with foci on the y-axis is ( ).
A. $\frac{x^2}{16}+\frac{y^2}{9}=1$
B. $\frac{y^2}{16}+\frac{x^2}{9}=1$
C. $\frac{x^2}{9}-\frac{y^2}{16}=1$
D. $\frac{y^2}{9}-\frac{x^2}{16}=1$

18. 抛物线$x^2=-12y$的焦点坐标是 ( ).The focal coordinates of the parabola $x^2=-12y$ are ( ).
A. $(3,0)$
B. $(-3,0)$
C. $(0,3)$
D. $(0,-3)$

19. 抛物线$y^2=8x$的焦点坐标是 ( ).The focal coordinates of the parabola $y^2=8x$

are (　).

A. $(2,0)$

B. $(-2,0)$

C. $(0,2)$

D. $(0,-2)$

二、填空题（Fill-in-the-Blank Questions）

1. $(x-3)^2+(y-4)^2=16$的圆心是______；半径是______.The center of $(x-3)^2+(y-4)^2=16$ is ______; the radius is ______.
2. $(x+5)^2+(y-2)^2=25$的圆心是______；半径是______.The center of $(x+5)^2+(y-2)^2=25$ is ______; the radius is ______.
3. $x^2+(y+4)^2=36$的圆心是______；半径是______.The center of $x^2+(y+4)^2=36$ is ______; the radius is ______.
4. $x^2+y^2+6x+10y-2=0$的圆心是______；半径是______.The center of $x^2+y^2+6x+10y-2=0$ is ______; the radius is ______.
5. $x^2+y^2-8x-4y-5=0$的圆心是______；半径是______.The center of $x^2+y^2-8x-4y-5=0$ is ______; the radius is ______.
6. 椭圆$\frac{x^2}{36}+\frac{y^2}{16}=1$的焦点是______; 顶点是______；离心率是______.The foci of the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$ are ______; the vertices are ______; the eccentricity is ______.
7. 椭圆$\frac{x^2}{225}+\frac{y^2}{49}=1$的长轴长是______；短轴长是______.The length of the major axis of the ellipse $\frac{x^2}{225}+\frac{y^2}{49}=1$ is ______; the length of the minor axis is ______.
8. 椭圆$\frac{x^2}{16}+\frac{y^2}{25}=1$的焦点是______；焦距是______；离心率是______.The foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ are ______; the focal length is ______; the eccentricity is ______.
9. 椭圆$\frac{x^2}{25}+\frac{y^2}{49}=1$的焦点是______；焦距是______；长轴长是______.The foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{49}=1$ are ______; the focal length is ______; the length of the major axis is ______.
10. 椭圆$\frac{y^2}{49}+\frac{x^2}{9}=1$的焦点是______；顶点是______；离心率是______.The foci of the ellipse $\frac{y^2}{49}+\frac{x^2}{9}=1$ are ______; the vertices are ______; the eccentricity is ______.
11. 双曲线$\frac{x^2}{25}-\frac{y^2}{16}=1$的焦点是______；离心率是______.The foci of the hyperbola $\frac{x^2}{25}-\frac{y^2}{16}=1$ are ______; the eccentricity is ______.
12. 双曲线$\frac{y^2}{16}-\frac{x^2}{36}=1$的焦点是______.The foci of the hyperbola $\frac{y^2}{16}-\frac{x^2}{36}=1$ are ______.
13. 双曲线$\frac{x^2}{64}-\frac{y^2}{9}=1$的焦点是______；渐近线是______.The foci of the hyperbola $\frac{x^2}{64}-\frac{y^2}{9}=1$ are ______; the asymptotes are ______.
14. 双曲线$\frac{x^2}{16}-\frac{y^2}{25}=1$的焦点是______；顶点是______.The foci of the hyperbola $\frac{x^2}{16}-\frac{y^2}{25}=1$ are ______; the vertices are ______.

15. 抛物线$y^2=-6x$的焦点坐标是_____；准线方程是_____.The focal coordinates of the parabola $y^2=-6x$ are _____; the directrix equation is _____.
16. 抛物线$x^2=6y$的焦点坐标是_____；准线方程是_____.The focal coordinates of the parabola $x^2=6y$ are _____; the directrix equation is _____.
17. 圆$(x-4)^2+(y+2)^2=9$的圆心是_____；半径是_____.The center of $(x-4)^2+(y+2)^2=9$ is _____; the radius is _____.
18. 圆$x^2+y^2-5x-8y-24=0$的圆心是_____；半径是_____.The center of $x^2+y^2-5x-8y-24=0$ is _____; the radius is _____.
19. 椭圆$\frac{x^2}{25}+\frac{y^2}{49}=1$的焦点是_____.The foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{49}=1$ are _____.
20. 椭圆$\frac{x^2}{18}+\frac{y^2}{8}=1$的焦距是_____.The focal length of the ellipse $\frac{x^2}{18}+\frac{y^2}{8}=1$ is _____.
21. 双曲线$\frac{x^2}{25}-\frac{y^2}{36}=1$的焦点在_____轴上.The foci of the hyperbola $\frac{x^2}{25}-\frac{y^2}{36}=1$ are on the _____ axis.
22. 双曲线$\frac{x^2}{36}-\frac{y^2}{16}=1$的焦点在_____轴上.The foci of the hyperbola $\frac{x^2}{36}-\frac{y^2}{16}=1$ are on the _____ axis.
23. 抛物线$y^2=10x$的开口方向是_____.The opening direction of the parabola $y^2=10x$ is _____.
24. 抛物线$x^2=-10y$的开口方向是_____；焦点是_____.The opening direction of the parabola $x^2=-10y$ is _____; the focus is _____.

三、解答题（Solution Questions）

1. 求以$C(2,5)$为圆心，半径是$\sqrt{10}$的圆的方程.Find the equation of the circle with center $C(2,5)$ and radius $\sqrt{10}$.

2. 圆的方程是$(x-5)^2+(y+3)^2=16$，求圆心和半径.The equation of the circle is $(x-5)^2+(y+3)^2=16$, find its center and radius.

3. 圆的一般方程是$x^2+y^2+8x-6y-11=0$，求圆心和半径.The general equation of the circle is $x^2+y^2+8x-6y-11=0$, find its center and radius.

4. 写出下列圆的标准方程：Write the standard equations of the following circles:(1) 圆心过$(3,-2)$，半径是 6；(1) Center at $(3,-2)$ with radius 6;(2) 圆心过$(0,5)$，半径是 3.(2) Center at $(0,5)$ with radius 3.

5. 求顶点在原点，焦点在 y 轴上，并且经过点$P(2,-4)$的抛物线方程.Find the equation of the parabola with vertex at the origin, foci on the y-axis, and passing through point $P(2,-4)$.